Uveal Tract (Uvea)

Images: 5.1, 5.27, 5.5, 5,17, 5.18, 5.21, 5.22, 5.26, 12.8, 5.29, 5.30, 12.15, 5.32, 5.35, 5.38

I. The uvea is the middle layer of the eye.

   A. It is composed of 3 regions

II. Iris

   A. Iris characteristics

      1. Iris root: peripheral termination of the iris

         a. Joins the iris to the ciliary body

         b. Thinnest area of the iris

         c. Clinical correlate: iridodialysis

            (1) Iris root tears from the ciliary body

2. Blood vessels

    a. Branch from the major circle of the iris.

    b. Radial blood vessels course from iris root to pupil margin

    c. Minor circle of the iris

    d. Iris capillaries are not fenestrated

4. Pupil:
    a. Function:
    b. Size:
    c. Miotic:
    d. Mydriatic:
    e. Sphincter muscle
    f. Dilator muscle:

5. Posterior iris epithelium

    a. Heavily pigmented cells

    b. Clinical correlate: Pupillary ruff

    c. Clinical correlate: Pigmentary dispersion syndrome

        (1) Pigment from the posterior iris is dispersed into the anterior chamber

III. Ciliary body

A. Characteristics

1. Triangular shape sagittally

2. Ring-shaped structure from iris root to ora serrata

3. Two parts

a. Pars plicata

b. Pars plana

(1) The zonular fibers

B. Ciliary muscle

1. Smooth muscle

2. Parasympathetic stimulation

3. Sympathetic stimulation has no clinically significant effect on the ciliary body.

C. Ciliary body blood vessels

1. Capillaries are fenestrated with wide lumens

D. Epithelium

1. Inner layer of cells are joined by zonula occludens.

2. Form part of the blood-aqueous barrier

3. Produces and secretes aqueous humor

E. Ciliary body function

1. Produce and secrete aqueous

2. Part of the blood-aqueous barrier

    a. Clinical correlate: Uveitis

3. Accommodation

IV. Choroid

A. Choroidal arteries are branches of the short posterior ciliary arteries

1. The blood vessels get smaller as they branch and get closer to the retina.

2. Choriocapillaris

    a. Single layer of anastomosing, fenestrated capillaries

    b. Densest in the macular area: it is the sole blood supply to the macular area.

3. Choroidal veins join in vortex pattern (vortex veins) in each quadrant of the eye

B. Bruch's membrane (basal lamina)

a. Innermost layer of choroid

b. Tight adhesion between Bruch's membrane and RPE

(1) Clinical correlate: drusen and age-related macular degeneration